



Drive-in Bytes

Falling prices and enhanced functionality, be it for watching DVD movies or burning CDs, have made CD-Writers and DVD-ROM drives a smarter buying decision as compared to an ordinary CD-ROM drive

The digital age has brought with it the need to store vast amounts of data, which has in turn created a market that witnesses the spawning of bigger and faster storage devices with alarming regularity. The humble floppy drive is as good as extinct, and the CD-ROM is treading on similar grounds, with faster and more feature-rich CD-Writers and DVD-ROMs all set to dominate the food chain. CD-Writers, for example, have reached a stage where they can burn 700 MB of data in less than three minutes, while DVDs not only promise an end to multiple-CD games and applications (and the irritant of 'disk-swapping' that invariably follows), but also bring with them the wonderful world of DVD movies—the concept of the home theatre was never a more likely proposition. Of course, all of this would mean little if the devices in question weren't accessible to us mortals. The good news is that thanks to a very competitive market and a demanding consumer, both CD-Writers and DVD-ROM drives are well within the bounds of even the most meagre of budgets. Buoyant with this knowledge, join us as we evaluate 11 CD-Writers and seven DVD-ROM drive. Which one will you take home to mom?

Digit Test Process

The qualifying criterion for the CD-Writers was that they should be able to burn a CD-R at 32x speed or above. For the DVD-ROM drives, this was 12x and above.

The CD-Writers and DVD-ROM drives were tested on an Intel Pentium 4 2 GHz PC with 256 MB DDR-RAM and a 40 GB ATA/100 7,200-rpm hard drive with Windows XP Professional as the OS. The drives in test were connected as secondary master. The CD-Writers and DVD-ROM drives were tested with DMA enabled for all IDE channels.

The CD-Writers under test were benchmarked for their performance in reading information off a CD and in writing different forms of data on to a CD-R and rewriteable CD (CD-RW). The media used was Princo CD-R 700 MB, capable of writing at 48x. The media used for the re-write and erase test was 16x 700 MB ASUS CD-RW.

Test methodology

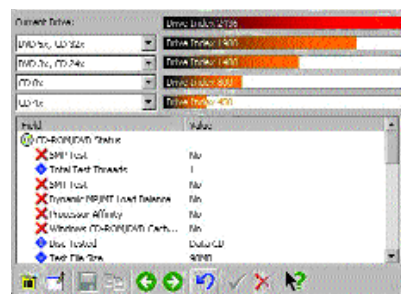
The drives were tested on four primary parameters: features (20 per cent weightage), performance (50 per cent weightage), value for money (20 per cent weightage) and warranty and support (10 per cent weightage).

Features: We checked for the buffer size, buffer underrun technology used by the CD-Writer, robustness of the CD tray, support for 8 cm CDs and whether the drive could be used vertically. We

also took into account the bundled software the nature and quality of the manuals and other package contents such as audio cable, IDE cable, software CDs and the number of blank CD-R and CD-Writer media provided. For DVD-ROM drives, we looked for the data buffer size, presence of audio and IDE cables and the quality of the documentation and software included.

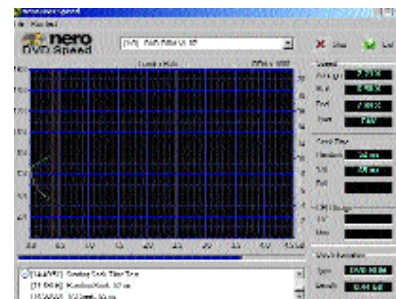
Performance: We looked at the writing, rewriting and reading performance of the CD-Writers. Here the scores that reflect the core performance of the Writer are the ones in the writing and rewriting tests. For the DVD-ROM drives, we evaluated the performance in terms of read speed and access time.

CD-Writers: We tested for ISO write speed by burning an ISO image created using Nero on the CD-R. Assorted data was written on a CD-R and a CD-RW using Nero 5.5.9.0. An Audio CD was created by burning WAV files on a CD-R. Our test CD was cloned using CloneCD 4.0 and we noted the time taken for this process. We also tested the time taken to erase the CD-RW. All tests were performed at the maximum speed supported. We checked whether the drive performed the operations successfully without errors during the writing process. Our read speed tests comprised the Nero CDSpeed v 1.0 test, which showed the average read speed of the CD-Writer. We also conducted assorted read tests where a combination of files ranging from 1 KB to 100 MB were read using the CD-Writer and copied to the hard disk—we noted the time taken for this entire process. The Writers were also benchmarked for sequential read, random read and average access time using SiSoft Sandra 2002 Pro. The disk registration time (time taken by the drive to display the label in the Explorer) was also noted.



SiSoft Sandra helps in determining the theoretical sequential speeds that a drive can achieve

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Nero DVD Speed is a great tool to check DVD drive speeds